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I. Data Source Design Overview

1. Reorganizing the Data

In the initial step, I reorganized the dataset. Tableau performs best when the data is structured in a flat format, meaning each row represents a unique record. To achieve this, I used Python for the data preparation process. Python was more efficient for handling the data, especially when working with larger datasets that exceed the threshold size in Tableau Public. By using Python, I was able to reorganize the data into a structure that Tableau could handle effectively.

2. Adding a Date Column

To enhance the realism of the analysis, I added a **Date** column by converting the **Day** values into actual dates. I assumed that the data spans from **January 5, 2022** to **September 30, 2024** (a 1000-day period). This transformation allowed for a more practical and realistic timeline, making it easier to analyze trends over time.

3. Adding the Admissions Column

Next, I added an **Admissions** column based on the provided formula:

$$\text{Inpatient Bed Count (current day)} = \text{Inpatient Bed Count (previous day)} + \text{Admissions (current day)} - \text{Discharges (current day)}$$

Since there was no previous day for the first record, I initialized the value as 0 to avoid errors in calculating admissions.

4. Handling Missing Values

Afterward, I checked for missing values. While the **Patient** and **ALC** tables had no missing data, I identified missing values in the **ED table** for the "ED Admit (no bed)" field at **Site #3** on two days. To address this, I used an imputation method and replaced the missing values with the average "ED Admit (no bed)" values for **Site #3**.

5. Standardizing Site Names for Table Connections

To facilitate the connection between tables, I standardized the site names in the **ALC table**. I created a new column called "**Standardized Site**" since the format of the site names in the **ALC table** differed from the formats in the **Inpatients** and **ED tables**. This standardization was crucial for ensuring a smooth join between the tables.

6. Discrepancy in the ALC Table

During the validation process, I identified a discrepancy in the **ALC table**. Specifically, I noticed that the **Total ALC Count** did not always match the sum of **ALC Counts** for **Sites 1, 2, and 3**. Initially, I assumed that the total might include **Sites 4 and 5**, but upon closer inspection, I discovered that the total was sometimes less than the sum for certain records.

Understanding the data collection process in more detail could help clarify this issue. There are a few possible explanations for the discrepancy: some data may have been **excluded** from the total due to missing information or failing to meet specific inclusion criteria. Another possibility is that

some records were **counted multiple times**—perhaps a patient was assigned to more than one designation. Although the catalog states that patients are assigned to a single designation at a time, it also notes that this policy may change. Lastly, there could have been **data entry errors**.

Initially, I explored the possibility of estimating the total for these records by comparing the differences across designations. However, the discrepancies were present in multiple designations, not just **Designation 1**, but also in **Designations 4, 5, and others**.

Given that, I decided to treat the total and sum **separately**. For the analysis, I focus on the more **detailed data** from Sites 1, 2, and 3, since granular data provides more accurate insights and allows for a deeper understanding of the specific contributions from each site.

II. Assumptions Made

1. Discharges in the Inpatient Bed Count formula: I assumed that "Discharges" in the formula is equivalent to **Separations**. Therefore, I used the **Separation** data in the formula to calculate the **Admissions**.

2. Funded Occupancy Rate: According to the data catalog, the **Funded Occupancy Rate** is defined as the proportion of total funded beds that are occupied by patients at midnight on the indicated day. However, since I did not have data for the **total bed counts**, I interpreted this as the percentage of **inpatient beds that are funded**, meaning **occupied funded beds over total inpatient bed count**.

3. ED Admits: I assumed that **ED Admits** from each site remain within that site. For example, **ED Admits** from **Site 4** are admitted to the inpatient ward of **Site 4**, and not transferred to other sites.

4. Date Range Assumption: I assumed that the data covers a period from **January 5, 2022** to **September 30, 2024** (1000 days).

III. Tableau Implementation

In the Tableau **Data Source** view, the tables are joined on two key fields: **Day Number** and **Site**. This ensures consistency in the data across tables. The standardized site names enabled a seamless connection between the **Inpatients**, **ED**, and **ALC tables**. Figure 1 and Figure 2 show the Data Source View:

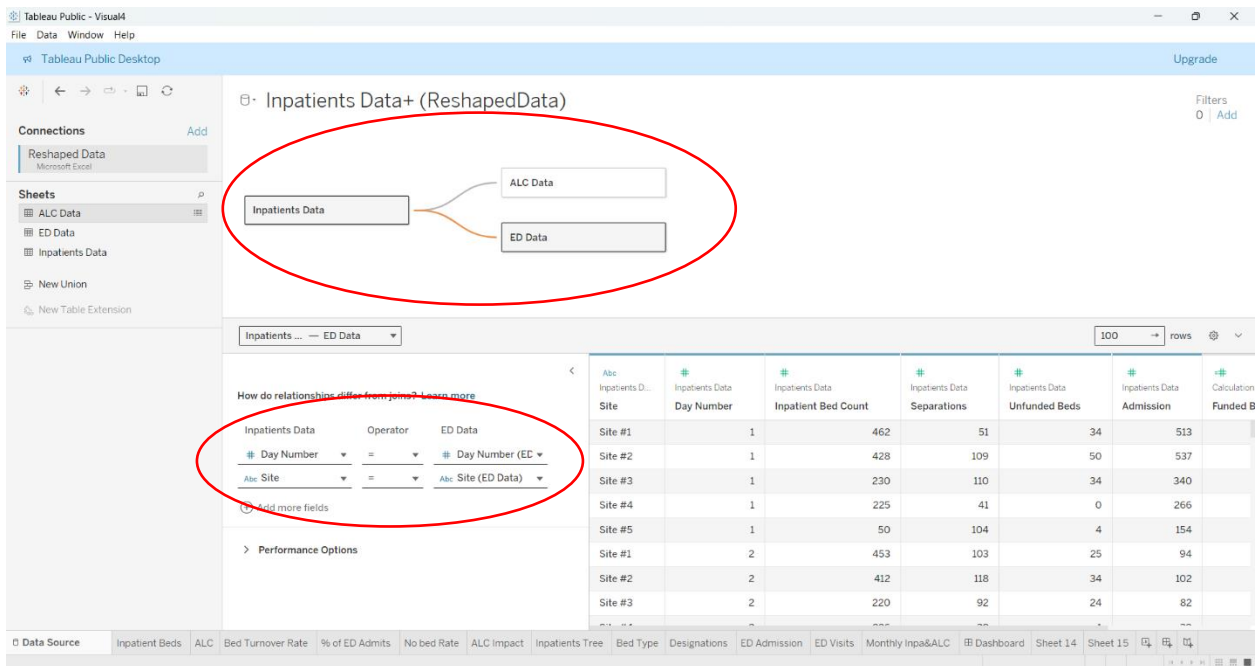


Figure 1. Joining Inpatients and ED tables

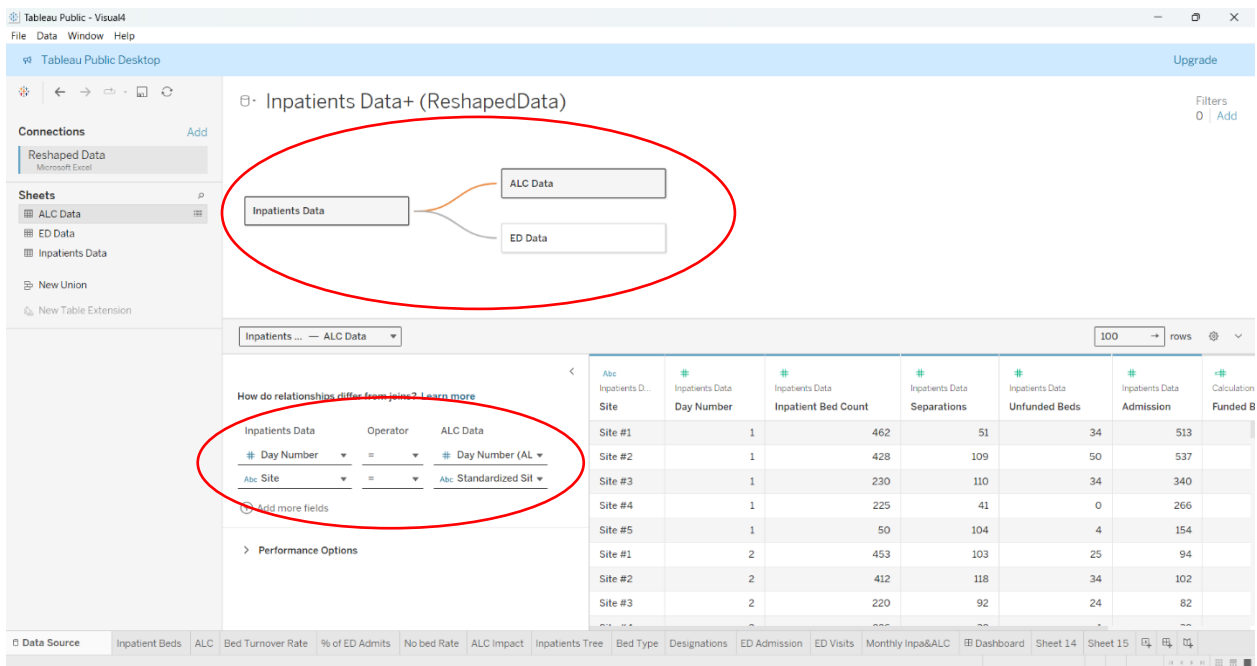


Figure 2. Joining Inpatients and ALC tables

The key calculated fields are also listed below:

Calculated Field	Description
Funded	The proportion of inpatient beds that are funded.
Unfunded	The proportion of inpatient beds that are unfunded.
Bed Turnover Rate	The ratio of patient separations to the total inpatient bed.
Patient Flow Rate	The ratio of patient separations to the number of admissions.
ED Admission Share	The proportion of admissions that come from ED.
No Bed Rate	The proportion of admitted patients without an immediately available inpatient bed.
ALC Impact Rate	The proportion of inpatient beds that are occupied by ALC patients.